
VASP Workflow

User Manual

Browser GUI · CLI Agents · SLURM HPC

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Contents

1	Introduction	4
1.1	Key features	4
1.2	Requirements	4
2	Installation	5
2.1	default_paths — platform configuration	5
3	Browser GUI	5
3.1	Starting the GUI	5
3.2	Three-tab workflow	6
3.3	Setup tab — field guide	6
3.3.1	Structure and POTCAR	6
3.3.2	Functional	6
3.3.3	Spin / SOC	6
3.3.4	Other flags	7
3.4	Task checkboxes	7
3.5	Band structure options	7
3.6	DOS options	7
3.7	Workflow tab	7
3.8	Results tab	8
4	CLI Agents	8
4.1	vasp-agent.py — workstation	8
4.2	vasp-agent-slurm.py — SLURM clusters	8
4.3	Remote / headless setup (no GUI access)	8
4.4	Generated directory structure	9
5	High-Throughput Screening (Materials Project)	9
5.1	Execution order	10
6	instructions.txt Format	10
6.1	Example	10

6.2	Keyword reference	12
7	Methods	12
7.1	Functionals	12
7.2	Spin-orbit coupling	13
7.3	Magnetic materials (collinear spin)	13
7.4	GGA+U	13
7.5	Automatic k-mesh generation	13
8	Advanced INCAR Control	14
8.1	Raw INCAR passthrough	14
8.2	Command-line INCAR overrides (<code>-incar</code>)	15
8.3	Constant-pressure relaxation	15
9	K-Path for Band Structure	15
9.1	Auto-detection (recommended)	15
9.2	Manual override	16
10	DOS Projections	16
11	ELF Bond Profiles	17
12	Convergence Tests	17
12.1	Shifted test structure	17
12.2	Automatic parameter selection	18
13	Parallelization	18
13.1	Auto-defaults	18
13.2	Manual overrides	18
14	SLURM Cluster Settings	18
14.1	profiles/slurm.json	18
14.2	Per-project overrides (in <code>instructions.txt</code>)	19
15	Wannier90	19
16	Born Charges, Dielectric & Piezoelectric Tensors (DFPT)	19
16.1	Example	20
17	Phonons (phonopy)	20
18	LOBSTER Bonding Analysis (COHP / COBI / COOP)	20
19	Worked Examples	21
19.1	MoS ₂ monolayer with SOC	21
19.2	Nickel: magnetism + GGA+U	21
19.3	GaAs: phonons with NAC	21
19.4	High-pressure relaxation with custom INCAR tags	22
19.5	Bi ₂ Se ₃ : topological insulator	22
20	Common OUTCAR Checks	23
21	Troubleshooting	23

22 Version History	24
Addendum: High-Throughput Quick Reference	25

1 Introduction

VASP Workflow is a framework for setting up, running, and analysing DFT calculations with VASP. Two interfaces are provided:

- **Browser GUI** (`vasp-gui.py`) — a Flask web application: configure calculations through a form, run steps from the browser, view band-structure and DOS plots in real time.
- **CLI agents** (`vasp-agent.py` for workstations, `vasp-agent-slurm.py` for SLURM clusters) — read a plain-text `instructions.txt` file and generate all INCAR, KPOINTS, POSCAR, POTCAR, and job scripts automatically.

Both interfaces share the same `modules/` back-end.

1.1 Key features

- Functionals: PBEsol, PBE, R2SCAN, HSE06, RVV10, LDA, AM05
- Spin-orbit coupling (SOC) and collinear spin (ISPIN=2)
- Magnetic materials: correct MAGMOM per atom; spin-resolved band and DOS plots (spin-up / spin-down)
- GGA+U (Dudarev, LDAUTYPE=2) per element and orbital
- Automatic k-path detection via spglib (Setyawan–Curtarolo standard paths), with a geometric fallback
- Convergence testing: ENCUT and k-mesh sweeps
- DOS with orbital projections via `sumo`
- Phonons via phonopy, Wannier90 interface, DFPT Born charges
- LOBSTER bonding analysis (COHP/COBI/COOP) via the `08_lobster` step
- Smart KPAR/NCORE auto-defaults scaled with MPI task count
- SLURM workflow with dependency-chained job submission

1.2 Requirements

Package	Notes
VASP 5.4+	<code>vasp_std</code> , <code>vasp_nc1</code> (SOC), optionally <code>vasp_gam</code>
POTCAR library	PAW pseudopotentials in per-element folders
Python 3.8+	Conda environment recommended
Flask	<code>pip install flask</code> — GUI only
<code>sumo</code> ≥ 2.0	<code>pip install sumo</code> — band and DOS plots
MPI	<code>mpirun</code> , <code>mpiexec</code> , or <code>srun</code>
phonopy	optional; <code>conda install -c conda-forge phonopy</code>
wannier90.x	optional; <code>conda install -c conda-forge wannier90</code>

2 Installation

```
# 1. Clone the repository
git clone https://github.com/KeivanS/VASP-Flow.git
cd VASP-Flow

# 2. Install Python dependencies
pip install flask sumo

# 3. Optional: phonons and Wannier support
conda install -c conda-forge phonopy wannier90

# 4. First-time configuration (creates default_paths from template)
make setup
nano default_paths      # set VASP paths, POTCAR directory, MPI
                        command
```

2.1 default_paths — platform configuration

The file `default_paths` (Makefile key=value syntax) is read at startup and its variables are exported as environment variables that the GUI uses as initial defaults. Settings saved via the GUI *System Setup* page take precedence; `default_paths` acts as the fallback for new installs or CLI-only use.

```
PYTHON          = python3
VASP_STD        = /path/to/vasp_std
VASP_NCL        = /path/to/vasp_ncl
VASP_GAM        = /path/to/vasp_gam      # optional
VASP_POTCAR_DIR = /path/to/potpaw_PBE
MPI_LAUNCH      = mpirun -np            # or: srun
MPI_NP          = 16
WANNIER90_X     = wannier90.x
LOBSTER_X       = lobster               # lobster binary or full
path
VISUALIZER_X    = ~/bin/jmol             # Jmol, VESTA, OVITO, ...
```

Three-layer configuration precedence (highest wins):

1. GUI System Setup page → saved to `settings.json`; also auto-synced to `profiles/slurm.json` for SLURM agents.
2. `default_paths` → Makefile bootstrap, read on every `make run`.
3. Built-in code defaults.

`profiles/workstation.json` is used only when running `vasp-agent.py --profile workstation` directly from the command line without the GUI.

3 Browser GUI

3.1 Starting the GUI

```
make run      # launches on http://localhost:5001
```

On a SLURM cluster — run on the login node, open on your laptop via SSH port-forwarding:

```
# On your laptop:
ssh -L 5001:localhost:5001 username@cluster

# On the cluster login node:
make run

# Open http://localhost:5001 in your laptop's browser
```

3.2 Three-tab workflow

1. **Setup** — configure the project, then click *Generate Workflow*.
2. **Workflow** — run each calculation step; live VASP output streams to the browser.
3. **Results** — band-structure and DOS plots; OUTCAR analysis.

To return to a previous project use the *Select project* dropdown. → **Workflow** jumps to the run tab; **Edit & Regenerate** re-opens Setup with all settings pre-filled.

3.3 Setup tab — field guide

3.3.1 Structure and POTCAR

Field	Description
Project name	Folder created in the working directory.
POSCAR	Paste the crystal structure, or click <i>Load file</i> . Click <i>Visualize</i> to convert the current POSCAR to XYZ and open it in the structure viewer configured under System Setup (Jmol, VESTA, OVITO, ...).
POTCAR directory	Path to your PAW library. If an element has multiple variants (Ga vs Ga_d) a chooser appears.

3.3.2 Functional

Choice	Description
PBESol	Optimised for solids; recommended starting point.
PBE	Standard GGA; widely used.
R2SCAN	Meta-GGA; good accuracy/cost balance.
HSE06	Hybrid — accurate band gaps, significantly slower.
RVV10	van der Waals; for layered or molecular materials.
LDA / AM05	Rarely needed.

3.3.3 Spin / SOC

Choice	Effect
None	Non-magnetic.
Collinear spin (ISPIN=2)	Sets ISPIN = 2; reveals the <i>Magnetic moment / atom</i> field (Section 7.3).
SOC z/x/y	Sets LSORBIT, LNONCOLLINEAR, SAXIS; uses the vasp_nc1 binary.

3.3.4 Other flags

Field	Effect
Hexagonal BZ	Correct M point for hexagonal structures.
2D / slab	Sets $k_z = 1$ (Gamma-only out-of-plane).
GGA+U	Adds Hubbard U rows per element/orbital (Section 7.4).
Execution profile	Select <i>SLURM HPC</i> to generate <code>sbatch</code> scripts; edit <code>profiles/slurm.json</code> first.
MPI tasks	CPU cores per calculation.

3.4 Task checkboxes

Task	Directory	Notes
Structure relaxation	01_relax	IBRION=2; run first if geometry is not pre-optimised.
SCF	02_scf	Required before bands or DOS.
Band structure	03_bands	Line-mode KPOINTS; Section 9.
DOS	04_dos	Dense k-mesh; orbital projections via sumo.
DFPT	06_dfpt	Born charges, dielectric + piezoelectric tensors; Section 16.
Phonons	07_phonons	Phonopy finite-displacement; DFPT needed for NAC.
Wannier90	05_wannier	NSCF interface; Section 15.
LOBSTER	08_lobster	COHP/COBI/COOP bonding analysis; needs SCF; Section 18.

Note. Normal execution order: **relaxation** → **SCF** → **bands** → **DOS**. Each step copies POSCAR/CHGCAR from the previous one automatically.

3.5 Band structure options

Leave the *k-path* field blank to trigger **automatic detection** from the POSCAR lattice vectors (Section 9). Enter a path manually to override, e.g. **G-M-K-G**.

3.6 DOS options

Add projection rows to request orbital-resolved DOS:

Element: Mo Orbitals: **d** Element: S Orbitals: **p**

For spin-polarised calculations sumo plots spin-up positive and spin-down mirrored below zero automatically.

3.7 Workflow tab

Each step shows three controls: **Run** (submit the step), **Edit files** (open INCAR/KPOINTS/-POSCAR in the browser; a `.bak` is created on first edit), and a **status badge**. **Run All Steps** chains every step and stops on failure.

3.8 Results tab

Panel	Contents
Band structure	Spin-polarised: blue = $\uparrow$, red dashed = $\downarrow$, same k-axis, $E_F = 0$. Non-magnetic: single-colour sumo plot. If sumo fails (e.g. a pymatgen gap-detection crash) the GUI falls back to a built-in <code>vasprun.xml</code> plotter so a band plot still renders.
Cumulative DOS	Stacked filled-area by element. Spin-polarised: up positive, down mirrored.
Projected DOS	Orbital-resolved via sumo; spin-down negated automatically.
OUTCAR analysis	Energy, Fermi level, magnetic moment, forces.

4 CLI Agents

4.1 vasp-agent.py — workstation

```
./vasp-agent.py
./vasp-agent.py -i my_inst.txt -s struct.vasp

cd ProjectName
./run_all.sh           # runs every step in sequence
./analyze.sh           # plots after all steps complete
```

4.2 vasp-agent-slurm.py — SLURM clusters

```
./vasp-agent-slurm.py           # uses profiles/slurm.json
./vasp-agent-slurm.py --profile slurm

cd ProjectName
./submit_all.sh                 # sbatch with dependency chaining
squeue -u $USER                 # monitor jobs
./analyze.sh                     # run locally after jobs finish
```

Two-phase workflow when convergence tests are requested:

```
./submit_convergence.sh         # STEP 1 -- submit convergence jobs
# review results, then:
./submit_calculations.sh        # STEP 2 -- prompts ENCUT/k-mesh, submits
```

4.3 Remote / headless setup (no GUI access)

When running on a remote supercomputer where the browser GUI is not available, executable paths and MPI settings must be configured directly in `profiles/slurm.json` (for the SLURM agent) or `profiles/workstation.json` (for the workstation agent).

```
# On the cluster, after cloning VASP-Flow:
nano profiles/slurm.json
```

The relevant keys are:

```
{
  "vasp_std":    "/path/to/vasp_std",
  "vasp_ncl":    "/path/to/vasp_ncl",
  "vasp_gam":    "/path/to/vasp_gam",
  "wannier90_x": "wannier90.x",
  "lobster_x":   "/path/to/lobster",
```

```

"mpi_cmd":      "srun",
"mpi_np":       128,
"modules":      ["intel/2023", "vasp/6.4"],
"slurm": {
  "partition":  "standard",
  "nodes":       2,
  "ntasks_per_node": 64,
  "time":        "12:00:00",
  "account":     "your-project"
}
}

```

The modules list is prepended as `module load ...` lines in every generated `run.sh`. Alternatively, all path keys can be set as environment variables in your `.bashrc` (`VASP_STD`, `LOBSTER_X`, etc.) and the profile values left as empty strings to inherit them.

Note: `profiles/slurm.json` is auto-regenerated whenever you click *Save Settings* in the GUI, so the authoritative source for GUI users is the System Setup page. On a headless cluster, edit the JSON file directly and leave it in place.

4.4 Generated directory structure

```

ProjectName/
  instructions.txt
  POSCAR
  POTCAR                      built once from $VASP_POTCAR_DIR
  01_relax/                   INCAR  KPOINTS  POSCAR  POTCAR  run.sh
  02_scf/
  03_bands/
  04_dos/
  08_lobster/                 if LOBSTER task selected (ISYM=0 NSCF +
    lobster)
  analysis/                   plots generated by analyze.sh
  run_all.sh                  workstation sequential runner
  submit_all.sh               SLURM job submission (dependency chained)
  analyze.sh                  post-processing: plots + OUTCAR extraction

```

5 High-Throughput Screening (Materials Project)

`ht-mp-scf.py` drives a batch of SCF + ELF calculations straight from Materials Project IDs. For every `mp-id` it downloads the **primitive** cell from MP, stages a per-material `instructions.txt` (an SCF task plus an INCAR `scf:` block setting `LELF = .TRUE.` for the electron localization function), and writes a `runall.sh` that calls the chosen agent to build the inputs and then runs or submits each job. Each SCF uses the automatic **coarse** density (1000 kpra, cell-aware) by default; override via `KMESH_DENSITY: fine` in `_ht_inputs/<id>/instructions.txt` or supply an explicit `--kmesh` grid.

```

pip install pymatgen mp-api          # MP download + primitive
                                     standardisation
export MP_API_KEY=...                 # Materials Project API key
export VASP_POTCAR_DIR=...            # POTCAR library (used by the
                                     agent)

# one mp-id per line (blank lines and # comments ignored)
printf 'mp-149\nmp-2534\n' > highthrouput_list

```

```
./ht-mp-scf.py --agent local --mpi 16 # workstation
./ht-mp-scf.py --agent slurm --profile slurm # cluster (chained by
    default)

./runall.sh # build inputs + run/submit
    every job
```

The ELFCAR for each material is written to <mp-id>/02_scf/. A summary `_ht_inputs/elements_Z.txt` tabulates every element in the screening set with its atomic number Z (from a built-in table — no MP query), plus a per-material breakdown.

Option	Meaning
<code>--agent</code>	<code>local</code> (vasp-agent.py) or <code>slurm</code> (vasp-agent-slurm.py); prompts if omitted.
<code>--list</code>	ID file; default <code>highthroughput_list</code> .
<code>--functional</code>	PBE (default), PBEsol, R2SCAN, ...
<code>--mpi</code>	MPI tasks per job.
<code>--encut</code>	Optional ENCUT override (eV).
<code>--kmesh</code>	Explicit SCF Γ -mesh, e.g. '8 8 8' or '12' (cubic); overrides the automatic density. Without this flag, the cell-aware coarse (1000 kpra) density is used. Per-material overrides via the <code>KMESH</code> or <code>KMESH_DENSITY</code> line in <code>instructions.txt</code> .
<code>--chain / --no-chain</code>	SLURM only; see below.

5.1 Execution order

Mode	Behaviour
<code>local</code>	<code>run_all.sh</code> blocks — materials run strictly one-at-a-time.
<code>slurm --chain</code> (default)	Each material's SCF is submitted with <code>--dependency=afterok</code> on the previous one, so the cluster runs them sequentially even though all are queued up front.
<code>slurm --no-chain</code>	Materials are submitted independently and run whenever the scheduler allows.

Note. The driver fetches structures over the network (Materials Project), so run it on a node with internet — typically the login node. `runall.sh` performs no downloads and can run on a compute node.

6 instructions.txt Format

6.1 Example

```
Project: My material name

Methods: PBEsol functional
         collinear magnetization
         GGA+U with U=4.0 on Fe-d orbitals

Tasks: structure relaxation
       SCF calculation
       band structure
       DOS (projected on Fe-d and O-p)

Convergence: Test k-points 6x6x6 to 12x12x12
```

```
ENCUT 400-600 eV  
  
MAGMOM: 4.0  
NSW: 100  
EDIFFG: -0.01  
MPI: 128  
NODES: 2  
NTASKS_PER_NODE: 64  
WALLTIME: 12:00:00
```

The parser is case-insensitive and searches the entire file, so section order is flexible.

6.2 Keyword reference

Keyword	Meaning
Project:	Project name; becomes the output folder.
MAGMOM:	Initial moment in μ_B /atom (collinear only). Written as MAGMOM = N*value . Default 2.0.
ISIF:	Relaxation type: 3=full, 2=ions only, 4=fixed volume.
NSW:	Max ionic steps. Default 100.
EDIFFG:	Force criterion in eV/Å. Default -0.01.
ENCUT:	Manual plane-wave cutoff in eV.
KMESH:	Explicit Gamma k-mesh overriding the automatic density, e.g. 8 8 8 or 12 (cubic). Applies to every automatic mesh (relax/SCF/DOS); for 2-D give the third value (12 12 1).
KMESH_DENSITY:	Automatic k-mesh density tier: coarse (1000 kpra, default for relaxation and convergence tests) or fine (5000 kpra, default for SCF/-DOS/spectroscopic steps). $kpra = N_k \times N_{atoms}$ where $N_k = N_x N_y N_z$; subdivisions scale as $N_i \propto \mathbf{b}_i^* $. Overridden by explicit KMESH .
PRESSURE:	External pressure for constant- P relaxation, e.g. 10 GPa or 50 kbar. Forces ISIF=3 , IBRION=2 , PSTRESS (Section 8.3).
INCAR: ... END_INCAR	Block of literal INCAR tags merged into the generated INCAR(s); optionally per-step (Section 8.1).
NKPTS:	k-points per band segment. Default 40.
MPI:	Total MPI tasks (nodes $\times$ cores/node).
KPAR:	Global k-point parallelisation.
NCORE:	Global cores-per-orbital.
SCF_KPAR: etc.	Per-step KPAR: also BANDS_, DOS_, RELAX_, PHONONS_ (DFPT is forced to KPAR = NCORE = 1).
SCF_NCORE: etc.	Per-step NCORE; same prefixes.
NODES:	Compute nodes (SLURM).
NTASKS_PER_NODE:	MPI tasks per node (SLURM).
PARTITION:	SLURM partition.
WALLTIME:	Time limit, e.g. 12:00:00.
ACCOUNT:	SLURM account / allocation.
DFPT_EDIFF:	EDIFF for DFPT. Default 1E-8.
PHONONS_DIM:	Supercell, e.g. 2 2 2.
PHONONS_MESH:	q-mesh, e.g. 20 20 20.
PHONONS_DISP:	Displacement in Å. Default 0.01.
PHONONS_BAND:	High-symmetry path for phonon bands.
PHONONS_NAC:	FALSE to disable NAC.
WANNIER_NUM_WANN:	Number of Wannier functions.
WANNIER_PROJ:	Projections, e.g. Ga:s;As:p.
WANNIER_EWIN:	Energy window, e.g. -5.0 to 15.0.

7 Methods

7.1 Functionals

```

Methods: PBEsol functional      # recommended for solids
Methods: R2SCAN functional      # meta-GGA
Methods: HSE06 functional       # hybrid -- more accurate but slower
Methods: RVV10 functional       # van der Waals

```

Default ENCUT values: PBE 400 eV, PBEsol 450 eV, R2SCAN 680 eV, HSE06 400 eV, RVV10 450 eV.

7.2 Spin-orbit coupling

Methods: SOC with magnetization in z-direction

Uses vasp_ncl. Sets LSORBIT, LNONCOLLINEAR, and SAXIS.

7.3 Magnetic materials (collinear spin)

Methods: collinear magnetization

MAGMOM: 4.0

Sets ISPIN = 2 and MAGMOM = N * value in every INCAR, where N is the atom count from the POSCAR.

Tip. Typical starting moments: Fe, Co $\approx 4\text{--}5 \mu_B$; Ni $\approx 2 \mu_B$; Cr, Mn $\approx 3\text{--}4 \mu_B$; rare earths $\approx 7 \mu_B$. VASP converges the self-consistent value automatically.

Antiferromagnets — after generation, edit INCAR to alternate signs:

MAGMOM = 4.0 -4.0 4.0 -4.0 ! alternating per sublattice

Spin-resolved plots produced automatically after DOS/bands run:

- *Cumulative DOS* — spin-up above the x-axis; spin-down mirrored below zero; elements colour-coded; solid = $\uparrow$, dashed = $\downarrow$.
- *Band structure* — spin-up in blue, spin-down in red dashed, same k-axis, $E_F = 0$.
- *Projected DOS* — sumo natively negates the spin-down channel in the orbital-resolved PDF.

Check the converged moment:

```
grep "magnetization_□(x)" OUTCAR | tail -5
```

7.4 GGA+U

Methods: GGA+U with U=4.0 on Fe-d orbitals

GGA+U with U=3.0 on Ni-d orbitals

GGA+U with U=6.0 on Ce-f orbitals

Dudarev scheme (LDAUTYPE=2); $U_{\text{eff}} = U - J$ with $J = 0$ set automatically. In the Dudarev formalism the parameter labelled U is already U_{eff} , so no separate J entry is needed. Combine freely with **collinear magnetization**.

The automatic lookup is triggered whenever the structure contains a transition-metal or rare-earth element *and* at least one electronegative anion from the set O, F, S, Se, Te, Cl, Br, I. This covers oxides, sulfides (e.g. AgCrP₂S₆), selenides, tellurides, and halides. U values are read from hubbard_u_defaults.csv; elements absent from the table (e.g. Ag with a filled d^{10} shell) receive no correction.

7.5 Automatic k-mesh generation

All Gamma-centred meshes are generated automatically from the POSCAR geometry. Two density tiers are available via the KMESH_DENSITY key (Section 6.2):

Tier	kpra	Typical use	Notes
coarse	1000	Relaxation, convergence tests	Agent k-point sweeps start here
fine	5000	SCF, DOS, spectroscopic steps	Default for all production steps

kpra (k-points per reciprocal atom) = $N_x N_y N_z \times N_{\text{atoms}}$. Given a target kpra and the reciprocal lattice vectors $\mathbf{b}_i^*$ (columns of $2\pi(\mathbf{A}^{-1})^\top$), the algorithm solves

$$N_i = \lceil \alpha |\mathbf{b}_i^*| \rceil_{\text{even}}, \quad \alpha = \left(\frac{N_k}{|\mathbf{b}_1^*| |\mathbf{b}_2^*| |\mathbf{b}_3^*|} \right)^{1/3}, \quad N_k = \text{kpra} / N_{\text{atoms}}.$$

Each N_i is rounded to the nearest even integer so that BZ boundary points ($k = 1/2$) are always included. For 2-D slabs N_z is fixed to 1.

Hexagonal cells ($a = b$, $\gamma = 120^\circ$): in-plane subdivisions are fixed at 6 (coarse) or 12 (fine); N_z is still computed from $|\mathbf{b}_3^*|/|\mathbf{b}_1^*|$ to maintain uniform sampling.

DOS mesh: always $2N_x \times 2N_y \times 2N_z$, ensuring denser sampling for accurate spectral features. The DFPT mesh is forced to match the SCF mesh exactly.

An explicit **KMESH**: `N1 N2 N3` entry overrides the automatic density for all steps.

8 Advanced INCAR Control

Any VASP keyword can be specified explicitly — nothing is hard-wired. Three routes, in increasing precedence: (1) an `INCAR: ... END_INCAR` block in the instructions file (global or per-step); (2) the agents' `-incar` command-line option; (3) direct editing — in the GUI every generated INCAR is editable per step (Workflow tab → *Edit files*, with an automatic `.bak` backup), and of course the files on disk can be edited before running. This section covers the first two plus **constant-pressure relaxation**.

8.1 Raw INCAR passthrough

Place literal INCAR lines — in ordinary VASP syntax — inside an `INCAR: ... END_INCAR` block in your `instructions.txt`. They are merged into the generated INCAR(s):

```
INCAR:                                # global block -- applies to every step
  LREAL  = Auto
  ALGO    = Fast
  NELMIN  = 6
END_INCAR
```

A tag that the agent already writes is **overwritten in place** (so `ALGO = Fast` above replaces the default `ALGO = Normal`); any tag the agent does not normally emit is appended under a clearly labelled `# User INCAR overrides` section, so nothing is silently lost.

To target a single step, name it on the header line. Recognised steps are `relax`, `scf`, `bands`, `dos`, `wannier`, `dfpt`, `phonons`, `lobster`:

```
INCAR scf:                            # applies only to 02_scf
  EDIFF  = 1E-8
  SIGMA  = 0.10
END_INCAR
```

Note. Precedence: a *per-step* block beats the *global* block, and both beat the agent's generated defaults. This is the recommended way to set tags the parser has no dedicated keyword for (e.g. `LMAXMIX`, `ADDGRID`, `LVTOT`, `IVDW`, `AMIX`). Full-line `#` comments in the instructions file are ignored by the parser, so a task keyword inside a comment does not enable that task.

8.2 Command-line INCAR overrides (-incar)

Both `vasp-agent.py` and `vasp-agent-slurm.py` accept a repeatable `-incar` option that injects explicit VASP tags at invocation time, without touching the instructions file:

```
./vasp-agent.py -i instructions.txt -s POSCAR \
  --incar "ALGO=Fast;NELMIN=6" \           # every step
  --incar "scf:NBANDS=64"                  # 02_scf only
```

Multiple tags in one option are separated by `;` (not commas — values such as `MAGMOM` may contain commas). An optional step prefix (`relax/scf/bands/dos/wannier/dfpt/phonons/lobster`) targets a single step. Command-line tags are applied *after* the instructions-file blocks, so they win on conflicts.

Note. Exceptions: physics-critical guards are re-imposed regardless of overrides — `ISYM=2` and `KPAR=NCORE=1` in `06_dfpt`, and the symmetry-off `ISYM` in `08_lobster`.

8.3 Constant-pressure relaxation

A single `PRESSURE` line turns the relaxation into a constant-pressure optimisation: the agent forces `ISIF = 3` (cell shape + volume + ions) and `IBRION = 2`, and imposes the target pressure through `PSTRESS`.

```
PRESSURE: 10 GPa          # also accepted: "5 kbar", "constant
  pressure 8"
```

VASP's `PSTRESS` tag is in **kBar** (1 GPa = 10 kBar). The agent converts automatically; if no unit is given, **GPa is assumed**. The resulting `01_relax/INCAR` reads, e.g.:

```
# Constant-pressure relaxation @ 10 GPa (PSTRESS = 100 kBar)
IBRION = 2
ISIF   = 3
NSW    = 100
EDIFFG = -0.01
PSTRESS = 100 ! external pressure in kBar (10 GPa)
```

Tip. `PSTRESS` adds the PV term so the relaxed cell sits at the requested external pressure. Use a tight `EDIFFG` and a well-converged `ENCUT`; the Pulay stress error grows at high pressure, so verify with a higher `ENCUT` before trusting the equilibrium volume.

9 K-Path for Band Structure

9.1 Auto-detection (recommended)

Omit the path in `instructions.txt` or leave the field blank in the GUI:

```
Tasks: band structure          # no "along ..." -> auto-detect from
  POSCAR
```

Since v1.6 the generator first determines the *actual space group* of the POSCAR with `spglib` (via `pymatgen`'s `KPathSetyawanCurtarolo`) and uses the standard Setyawan–Curtarolo high-symmetry path for that space group. Every high-symmetry point is transformed through Cartesian reciprocal space into the reciprocal basis of *this* POSCAR, so the path is correct for the actual cell used in the SCF regardless of orientation or setting.

If `spglib` or `pymatgen` is not installed (or the symmetry analysis fails), the generator falls back to the geometric classifier below: it reads the POSCAR lattice vectors, computes cell lengths

and angles, and selects a default path by Bravais type. 2-D structures always use the geometric route so the in-plane path is preserved.

Type	Detection criterion	Default path
cF (FCC)	$a=b=c$, angles $\approx 60^\circ$	G-X-W-K-G-L-U-W
cI (BCC)	$a=b=c$, angles $\approx 109.5^\circ$	G-H-N-G-P-H
cP (cubic)	$a=b=c$, 90°	G-X-M-G-R-X
hP (hexagonal)	$a \approx b$, $\gamma = 120^\circ$	G-M-K-G-A-L-H-A
tP (tetragonal)	$a \approx b \neq c$, 90°	G-X-M-G-Z-R-A-Z
oP (orthorhombic)	90° , $a \neq b \neq c$	G-X-S-Y-G-Z-U-R-T-Z
rP (rhombohedral)	$a=b=c$, equal angles	G-T-F-G-L
mP (monoclinic)	$\alpha = \gamma = 90^\circ$, $\beta \neq 90^\circ$	G-Y-A-Z-G-B-D

Note. For **conventional** FCC or BCC cells (right angles, 4 or 2 atoms), auto-detection returns the cubic (cP) path. Use the **primitive** cell for an unfolded FCC/BCC band structure — the primitive FCC cell has angles $\approx 60^\circ$ and is recognised automatically.

The detection route appears in the KPOINTS comment:

```
Line-mode KPOINTS [spglib ... -- auto-detected] # spglib route
Line-mode KPOINTS [FCC -- auto-detected]       # geometric
fallback
```

9.2 Manual override

Tasks: band structure along G-X-W-K-G-L

Label	Coords	Use
G	(0, 0, 0)	all Γ
M	(0.5, 0.5, 0)	cubic, tet (0.5, 0, 0) in hexagonal
K	(1/3, 1/3, 0)	hexagonal
R	(0.5, 0.5, 0.5)	cubic
Z	(0, 0, 0.5)	general
A	(0, 0, 0.5)	hex, tet
L	(0.5, 0, 0.5)	hexagonal
H	(1/3, 1/3, 0.5)	hexagonal
S	(0.5, 0.5, 0)	orthorhombic
W	(0.5, 0.25, 0.75)	FCC primitive
P	(0.25, 0.25, 0.25)	BCC primitive
N	(0, 0, 0.5)	BCC primitive

10 DOS Projections

All of the following forms are accepted:

```
DOS (projected on Mo-d and S-p)
DOS with projections on Ni:d and Ni:s
DOS with projection on Fe-d and O-p
```

Orbital channels: s, p, d, f. Separator is either - or :. Sumo generates an orbital-resolved PDF in analysis/.

11 ELF Bond Profiles

If an SCF calculation writes an **ELFCAR** (electron localization function, requested with **LELF = .TRUE.** — the high-throughput driver sets this automatically), the utility **elf_bonds.py** plots ELF along each non-equivalent nearest-neighbour bond, sampled from one atom centre to the other:

```
./elf_bonds.py [ELFCAR] [--out PREFIX] [--nn-scale 1.2] [--npoints 200]
```

It parses the **ELFCAR** grid directly (only numpy + matplotlib), groups bonds by element pair and length (rounded to 0.01 Å) to pick the non-equivalent ones, and interpolates ELF along each bond with periodic boundary conditions. Output is **FORMULA_elf_bonds.png/pdf** (the prefix defaults to the reduced formula, e.g. **MoSe2_elf_bonds.pdf**; override with **-out**) showing ELF vs. distance along the bond, with the ELF = 0.5 homogeneous-electron-gas reference marked. **analyze.sh** runs it automatically when **02_scf/ELFCAR** is present, and the GUI Results tab shows the same plot whenever the SCF wrote an **ELFCAR**.

Tip. Reading the profile: ELF is high (near 1) where electrons are paired and localized. A covalent bond therefore shows an ELF *maximum* in the bonding region *between* the atoms (e.g. Si-Si ≈ 0.95 at mid-bond), not at the nuclei. Because VASP builds the **ELFCAR** from the pseudo (valence) density, the value *at* an atom centre is a PAW artifact — often low, even ≈ 0 for heavy atoms — so read the bonding region, not the endpoints.

Note. VASP does *not* compute ELF for non-collinear / spin-orbit runs (LSORBIT/LNONCOLLINEAR); it prints "WARNING: ELF not implemented for non collinear case" and writes an all-zero **ELFCAR**. **elf_bonds.py** detects this and reports it rather than producing a blank plot — re-run the SCF *without* SOC to get ELF.

12 Convergence Tests

```
# Range (auto-steps)
Convergence: Test k-points from 6x6x1 to 18x18x1      # 2D material
              Test k-points from 6x6x6 to 12x12x12    # 3D material
ENCUT 400-600 eV
```

```
# Explicit mesh list
Convergence: Test k-points 6x6x3, 12x12x6, 18x18x9
```

Results in **00_convergence/kpoints/** and **00_convergence/encut/**.

```
./run_convergence.sh          # workstation
./submit_convergence.sh      # SLURM
./analyze_convergence.sh     # plots: energy / pressure / forces
```

12.1 Shifted test structure

The POSCAR used in **00_convergence** has **atom 1 displaced by (0.01, 0.02, 0.03) Å (Cartesian)** so that the force on it is non-zero and its convergence can be tracked — at the equilibrium (high-symmetry) positions the forces would vanish identically and the force panel would be meaningless. The displacement is converted through the lattice for Direct-mode POSCARs. All production steps (**01_relax**, **02_scf**, ...) use the *original*, *unshifted* positions.

12.2 Automatic parameter selection

After the runs, `00_convergence/choose_params.py` (invoked by `run_convergence.sh`, `analyze_convergence` and the phase-2 scripts) selects the smallest ENCUT and k-mesh from which every subsequent change stays below

$$\max_i |\Delta F_i^{\text{atom } 1}| < 5 \text{ meV/\AA} \quad \text{and} \quad \max_i |\Delta \sigma_{ii}| < 0.5 \text{ kbar},$$

and writes the result to `00_convergence/converged_params.txt`. Phase 2 (`run_calculations.sh`, `submit_calculations.sh`, and the GUI form) offers these values as defaults; when run *non-interactively* — e.g. a convergence test automatically followed by the SCF in a chained/nohup run — they are applied without prompting. If the criteria are not met within the tested range, the largest tested value is used and a NOT CONVERGED warning is issued. Thresholds can be overridden: `choose_params.py -ftol 3 -ptol 0.2`.

13 Parallelization

13.1 Auto-defaults

When KPAR/NCORE are not specified the agent computes from total tasks N :

Step	KPAR	NCORE	Notes
SCF / DOS / relax	$\lfloor \sqrt{N} \rfloor$	N/KPAR	
bands	1	N	Required for line-mode
DFPT	1	1	Forced: linear response / PEAD supports neither
phonons	1	$\lfloor \sqrt{N} \rfloor$	

Note. $\text{KPAR} \times \text{NCORE}$ must divide N evenly. The agent rounds down automatically to the nearest valid value.

13.2 Manual overrides

```
MPI: 128          KPAR: 4          NCORE: 8

# Per-step (take priority over globals)
SCF_KPAR: 4       BANDS_KPAR: 1     DOS_KPAR: 8
SCF_NCORE: 8      BANDS_NCORE: 16    DOS_NCORE: 8
RELAX_KPAR: 2     PHONONS_KPAR: 1    # DFPT is always KPAR=1, NCORE=1
```

14 SLURM Cluster Settings

14.1 profiles/slurm.json

```
{
  "vasp_std": "vasp_std",  "vasp_ncl": "vasp_ncl",
  "mpi_cmd":  "srun",
  "modules":  ["intel/2021", "impi/2021", "vasp/6.3.2"],
  "slurm": {
    "partition": "standard",
    "nodes":     2,
    "ntasks_per_node": 64,
    "time":      "12:00:00",
    "account":   "mygroup"
  }
}
```

}

14.2 Per-project overrides (in instructions.txt)

```
NODES: 4      NTASKS_PER_NODE: 32      PARTITION: gpu
WALLTIME: 48:00:00      ACCOUNT: myallocation
```

15 Wannier90

```
Tasks: Wannier90 wannierization

WANNIER_NUM_WANN: 8
WANNIER_PROJ: Ga:s;As:p
WANNIER_EWIN: -5.0 to 15.0
```

The agent generates `05_wannier/wannier90.win` with the NSCF k-mesh and projection block. After the NSCF run, execute `wannier90.x` from the `05_wannier/` directory.

16 Born Charges, Dielectric & Piezoelectric Tensors (DFPT)

The DFPT task creates `06_dfpt`, which computes Born effective charges Z^* , the macroscopic dielectric tensor, and the piezoelectric stress tensor from the converged SCF charge density.

Two routes are generated automatically depending on the method:

Method	Route
PBE, PBEsol, LDA, AM05 (no SOC)	Linear response: <code>IBRION=8, LEPSILON=.TRUE.</code>
SOC, HSE06, or R2SCAN	Finite field (PEAD): <code>IBRION=6, LCALCEPS=.TRUE.</code> — VASP's DFPT is not implemented for noncollinear magnetism, hybrids, or meta-GGAs. Requires an insulating system.

Note. Both routes support neither k-point nor band parallelisation: **KPAR** and **NCORE** are forced to 1 in the generated INCAR (overriding any global or per-step setting). The full MPI task count is still used for plane-wave parallelisation. Likewise **ISYM=2** is enforced (symmetry imposed): perturbation theory relies on the symmetry-reduced perturbation set, so a symmetry-off `ISYM=0/-1` — e.g. from a global INCAR: override block intended for the LOBSTER/NSCF steps — is never inherited by `06_dfpt`.

`copy_from_scf.sh` (invoked by `run.sh`) pulls `WAVECAR` and `CHGCAR` from `02_scf` at run time. If `02_scf` has been converted in place for LOBSTER (`ISYM=0/-1` in its INCAR, as done by the high-throughput `run_scf_for_lobster.sh`), the `WAVECAR` copy is skipped — it holds the full, unsymmetrised k-mesh, which the `ISYM=2` run cannot restart from — and VASP regenerates the orbitals from the `CHGCAR`, which the LOBSTER conversion (a fixed-charge NSCF with `LCHARG=.FALSE.`) leaves untouched. If the symmetry-off INCAR is *not* a fixed-charge NSCF, a warning is printed that the `CHGCAR` itself may stem from a symmetry-off SCF and a symmetric SCF should be re-run first.

`run.sh` executes `extract_born.py` after VASP, which writes:

- `BORN` — phonopy NAC format: electronic $\varepsilon_\infty + Z^*$ per atom.
- `born_charges.txt` — human-readable summary shown in the GUI Results tab: electronic (ion-clamped) ε_∞ , ionic contribution, total static $\varepsilon_0 = \varepsilon_\infty + \varepsilon_{\text{ion}}$; Z^* tensors labelled by element; an acoustic-sum-rule check ($\sum_{\text{ions}} Z^*$ should vanish — a large residual indicates

an under-converged k-mesh or EDIFF); and the piezoelectric stress tensors e (C/m², clamped-ion, ionic, and relaxed-ion total; meaningful only for non-centrosymmetric crystals).

DFPT_EDIFF in `instructions.txt` tightens the electronic convergence (default 1E-8).

16.1 Example

```
Project: GaAs
Methods: PBE functional
Tasks: relaxation, SCF, DFPT Born charges and dielectric
DFPT_EDIFF: 1E-8
MPI: 16
```

In the GUI, tick **Born Charges & Dielectric Constant (DFPT)** on the Setup tab (the DFPT EDIFF field is optional); after running `06_dfpt` from the Workflow tab, the full summary appears in the **Born charges** panel of the Results tab. With SOC, HSE06 or R2SCAN selected under Methods, the finite-field INCAR is generated automatically — no extra input is needed. For LO-TO splitting in phonon spectra, combine with the phonon task as in the GaAs worked example (Section 19.3).

17 Phonons (phonopy)

```
Tasks: phonons lattice dynamics

PHONONS_DIM: 2 2 2
PHONONS_MESH: 20 20 20
PHONONS_DISP: 0.01
PHONONS_BAND: G M K G A
PHONONS_NAC: FALSE      # disable non-analytic correction
```

Note. For the non-analytic correction (important for polar materials), include DFPT in your task list so Born effective charges are computed.

18 LOBSTER Bonding Analysis (COHP / COBI / COOP)

Adding LOBSTER COHP/COBI analysis (keywords: `lobster`, `cohpp`, `cobi`, `coop`, `bonding`) to the task list creates `08_lobster`: a symmetry-off NSCF that reads `02_scf/CHGCAR`, followed by a run of the `lobster` binary.

- **ISYM=0 is required** — a symmetrised k-mesh gives unusable COHP/ICOHP values (wrong sign, errors above 100%). The generated INCAR sets it automatically. With SOC enabled, **ISYM=-1** is used instead: spin-orbit coupling breaks time-reversal symmetry, which ISYM=0 would still assume.
- **NBANDS** is auto-set to at least the LOBSTER basis-function count; **LMAXMIX** is chosen from the elements (6 for *f*, 4 for *d*, 2 for *sp*).
- The LOBSTER binary is set with **LOBSTER_X** in `site.env` (default `lobster-5.1.0-OSX`); it can be overridden at runtime via `$LOBSTER_BIN`.

```
Project: NaCl
Methods: PBE functional
Tasks: SCF, LOBSTER COHP/COBI analysis
MPI: 16
```

Run with `./vasp-agent.py` (builds and runs `02_scf + 08_lobster`); the GUI shows COHP/COBI/COOP plots on the Results tab. To aggregate averaged COHP/COBI over many projects into `lobster_acohp_acobi.csv`, run `python3 lobster_postprocess.py`.

19 Worked Examples

19.1 MoS₂ monolayer with SOC

```
Project: MoS2 monolayer

Methods: PBEsol functional
         SOC with magnetization in z-direction

Tasks: structure relaxation
       SCF calculation
       band structure
       DOS (projected on Mo-d and S-p)

Convergence: Test k-points 8x8 to 20x20
             ENCUT 400-600 eV

MPI: 64
```

The generator detects hexagonal (hP) and uses G-M-K-G-A-L-H-A.

19.2 Nickel: magnetism + GGA+U

```
Project: Nickel GGA+U

Methods: PBEsol functional
         collinear magnetization
         GGA+U with U=4.0 on Ni-d orbitals

Tasks: structure relaxation
       SCF calculation
       band structure
       DOS (projected on Ni-d and Ni-s)

MAGMOM: 2.0
MPI: 128
NODES: 2
NTASKS_PER_NODE: 64
WALLTIME: 08:00:00
```

With a primitive FCC POSCAR (angles $\approx 60^\circ$) the generator detects cF and uses G-X-W-K-G-L-U-W. Band and DOS plots show both spin channels.

19.3 GaAs: phonons with NAC

```
Project: GaAs phonons
```

```

Methods: PBEsol functional

Tasks: SCF calculation
       DFPT Born charges and dielectric
       phonons lattice dynamics

PHONONS_DIM: 4 4 4
PHONONS_MESH: 20 20 20
PHONONS_BAND: G X W L G K

MPI: 64      WALLTIME: 24:00:00

```

19.4 High-pressure relaxation with custom INCAR tags

A constant-pressure cell relaxation at 10 GPa, combined with raw INCAR passthrough to tighten convergence and add tags the parser has no dedicated keyword for:

```

Project: Silicon 10GPa

Methods: PBEsol functional

Tasks: structure relaxation
       SCF calculation
       band structure

PRESSURE: 10 GPa      # constant-P relax -> ISIF=3, IBRION=2,
PSTRESS=100

# Tags applied to every step
INCAR:
  LREAL  = .FALSE.
  ADDGRID = .TRUE.
  LMAXMIX = 4
END_INCAR

# Tighter electronic convergence for the relaxation only
INCAR relax:
  EDIFF  = 1E-8
  EDIFFG = -0.005
END_INCAR

ENCUT: 600
MPI: 64

```

The 01_relax/INCAR is generated with ISIF=3, PSTRESS=100, the global tags appended, and EDIFF/EDIFFG overridden to the per-step values; 02_scf and 03_bands receive only the global block.

19.5 Bi₂Se₃: topological insulator

```

Project: Bi2Se3 topological

Methods: PBEsol functional
         SOC with magnetization in z-direction

Tasks: structure relaxation
       SCF calculation

```

```

    band structure
    DOS (projected on Bi-p and Se-p)
    Wannier90 wannierization

Convergence: Test k-points from 8x8x4 to 16x16x8
              ENCUT 400-600 eV

WANNIER_NUM_WANN: 18
WANNIER_PROJ: Bi:p;Se:p
WANNIER_EWIN: -2.0 to 2.0
MPI: 128

```

20 Common OUTCAR Checks

```

grep "reached_required_accuracy" OUTCAR          # converged?
grep "energy_ without"           OUTCAR | tail -1 # total energy
grep "E-fermi"                   OUTCAR | tail -1 # Fermi level
grep "magnetization_(x)"         OUTCAR | tail -5 # moment
grep "TOTAL-FORCE"               OUTCAR | tail -20
grep -A3 "BORN_EFFECTIVE_CHARGES" OUTCAR | head -20

```

21 Troubleshooting

Symptom	Solution
POTCAR not found	Check VASP_POTCAR_DIR in <code>site.env</code> or GUI System Setup. Use the POTCAR chooser for unusual variants (Ga_d, Fe_sv).
vasp_std not found	Set the full binary path in <code>site.env</code> or GUI System Setup.
WARNING: may not have converged	Increase NSW (ionic) or NELM (electronic) in INCAR and re-run.
Wrong bands ($W = \Gamma$)	Conventional cubic cell detected; use the <i>primitive</i> cell for FCC/BCC so auto-detection picks cF/cI.
Projected DOS missing	Check <code>instructions.txt</code> : <code>projected on</code> and <code>projections on</code> are both accepted. Verify element names match POSCAR.
KPAR error	$KPAR \times NCORE$ must divide total MPI tasks. Use auto-defaults or set explicitly valid values.
GUI unreachable	Check SSH tunnel: <code>ssh -L 5001:localhost:5001 user@cluster</code> .
Project missing from dropdown	Folder must contain POSCAR + <code>settings.json</code> in the directory where <code>vasp-gui.py</code> was started.
Band plot empty or not shown	Usually <code>sumo-bandplot</code> crashed on this band structure; the GUI now falls back to a built-in plotter automatically. An empty plot also means a wrong E_F : the Fermi level is read from 04_dos/OUTCAR (then 02_scf/OUTCAR), since 03_bands eigenvalues share the NSCF/DOS energy reference.

22 Version History

- v1.0 (2026-03-23)** Initial release — GUI, workstation agent, PBEsol/R2SCAN/HSE06, SOC, GGA+U, convergence tests, Wannier90, phonopy.
- v1.1 (2026-04-05)** Per-task KPAR/NCORE smart defaults; SLURM agent with dependency-chained submission; cumulative DOS by element (no pymatgen); copy-from-relax/scf scripts.
- v1.2 (2026-06)** Collinear magnetism: MAGMOM per atom, spin-resolved band and DOS plots. Auto k-path detection (8 crystal systems). Fixed DOS projection parsing. Separate GUI and agent quick-reference sheets. Rewritten comprehensive LaTeX manual.
- v1.3 (2026-06-15)** Raw INCAR passthrough (INCAR: ... END_INCAR, global or per-step) for arbitrary VASP tags. Constant-pressure relaxation via PRESSURE (ISIF=3, IBRION=2, PSTRESS).
- v1.4 (2026-06-19)** High-throughput driver `ht-mp-scf.py`: Materials Project primitive cells → batch SCF + ELF, with per-material element/ Z table and cross-material `afterok` chaining on SLURM.
- v1.5 (2026-06-24)** Robust band plotting across the GUI and both CLI agents (workstation + SLURM): the built-in `vasprun.xml` plotter now handles non-spin as well as spin, and is used automatically when `sumo-bandplot` crashes or is unavailable. Fixed the matplotlib fallback (`plot_results.py`) to resolve the real Fermi level and emit PNG, not just PDF.
- v1.6 (2026-07-08)** Band k-path derived from the real space group via `spglib` (Setyawan–Curtarolo) in all agents and the GUI, with the geometric classifier kept as fallback. New `08_lobster` step: LOBSTER COHP/COBI/COOP bonding analysis with GUI plots and the `lobster_postprocess.py` aggregator; the NSCF uses ISYM=0, or ISYM=-1 when SOC is on. Full GPL-3.0 text in LICENSE (GitHub now detects the license correctly). Manual and quick-reference sheets brought up to date.
- v1.7 (2026-07-08)** DFPT step overhaul: KPAR/NCORE forced to 1 (linear response and PEAD support neither); automatic finite-field route (LCALCEPS, IBRION=6) when SOC, HSE06 or R2SCAN is requested; `extract_born.py` now reports ϵ_∞ , the ionic contribution, total static ϵ_0 , an acoustic-sum-rule check, element-labelled Z^* , and clamped-/relaxed-ion piezoelectric tensors.
- v1.8 (2026-07-09)** ELF bond profiles: `elf_bonds.py` plots the electron localization function along non-equivalent nearest-neighbour bonds (atom centre to atom centre), wired into `analyze.sh` for both CLI agents and shown in the GUI Results tab whenever `02_scf/ELFCAR` exists. Output is named after the reduced formula (`MoSe2_elf_bonds.pdf`). Detects the all-zero ELFCAR that VASP writes for non-collinear / spin-orbit runs and reports it instead of emitting a blank plot.
- v1.9 (2026-07-22)** *K-mesh and GGA+U*. Physics-based cell-aware k-mesh generation replacing fixed uniform grids. Two density tiers: **coarse** (1000 k-points per reciprocal atom, kpra) for convergence tests and relaxations; **fine** (5000 kpra) for SCF, DOS, and spectroscopic steps. Subdivisions $N_i \propto |\mathbf{b}_i^*|$ from reciprocal lattice vectors, all-even to include BZ boundary points; $N_z = 1$ for 2-D slabs. Hexagonal systems: fixed 6×6 (coarse) or 12×12 (fine) in-plane, N_z from $|\mathbf{b}_3^*|/|\mathbf{b}_1^*|$. DOS mesh always $2N_x \times 2N_y \times 2N_z$. DFPT forced to match SCF mesh exactly. New instruction key `KMESH_DENSITY` (coarse/fine). GGA+U anion gate extended to all chalcogenides and halides (O, F, S, Se, Te, Cl, Br, I); Hubbard lookup table now covers sulfides such as AgCrP_2S_6 .

Plot improvements. ELF bond-profile and LOBSTER (COHP/COBI/COOP) plot legends and statistics boxes moved outside the axes area so they never obscure curves. Shared plotting module `modules/cohp_plot.py` used by both the GUI and the CLI agents; COHP, COBI, and COOP plots are now generated automatically in `analyze.sh` for both the workstation and SLURM agents, matching GUI behaviour.

Structure visualizer. New *Visualize* button in the POSCAR card: converts the current POSCAR to XYZ via `modules/poscar2xyz.py` and opens it in Jmol, VESTA, or any other viewer. The viewer path is set under System Setup → *Structure visualizer* (or `VISUALIZER_X` in `default_paths`).

Configuration. `LOBSTER_X` and `VISUALIZER_X` are now exposed in the GUI System Setup page and propagated to the agent subprocess environment. Platform config file renamed from `site.env` to `default_paths` for clarity; three-layer precedence (`default_paths` < GUI settings < profile JSON) documented. Auxiliary Python scripts (`band_plot.py`, `dos_plot.py`, `elf_bonds.py`, `lobster_postprocess.py`, `conv_choose.py`, `poscar2xyz.py`) consolidated into `modules/`.

Addendum: High-Throughput Quick Reference

Cheat-sheet for `ht-mp-scf.py` (full description in Section 5). For each Materials Project ID it downloads the **primitive** cell, stages an SCF calculation that also writes the electron localization function (`LELF = .TRUE.` → `ELFCAR`) on a cell-aware coarse Γ -mesh (1000 kpra default; override with `KMESH_DENSITY` or `--kmesh`), and builds a `runall.sh`.

```

pip install pymatgen mp-api
export MP_API_KEY=...           # Materials Project key
export VASP_POTCAR_DIR=...      # POTCAR library (used by the agent)

printf 'mp-149\nmp-2534\n' > highthrouput_list  # one mp-id per line

./ht-mp-scf.py --agent local    --mpi 16        # workstation,
sequential
./ht-mp-scf.py --agent slurm    --profile slurm  # cluster,
afterok-chained
./ht-mp-scf.py --agent slurm    --no-chain      # cluster,
independent

./runall.sh                     # build inputs + run/submit each SCF

```

Output	Location
ELFCAR (electron localization)	<mp-id>/02_scf/ELFCAR
Staged inputs (POSCAR + instructions)	_ht_inputs/<mp-id>/
Element / Z table	_ht_inputs/elements_Z.txt

Note. Transfer to a workstation: only the project folder is needed (code, modules/, profiles/); VASP binaries and the POTCAR library already live there. `ht-mp-scf.py` needs internet for the MP download, so run it on the login node — `runall.sh` does none.

Note. If the download step errors with 'dict' object has no attribute 'number': that is an spglib / pymatgen version mismatch. Fix the root cause with `pip install -U spglib`. The driver also falls back to `Structure.get_primitive_structure()` automatically, so a run is never blocked by it.

Source: <https://github.com/KeivanS/VASP-Flow> Compiled: July 22, 2026 Maintained by Keivan Esfarjani